VIVEK SHARMA
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EDUCATION

Ph.D. Mechanical Engineering and Applied Mechanics — **Anticipated May 2026**
University of Pennsylvania — Cumulative GPA: 3.8/4
M.S. Mechanical Engineering — **Graduated May 2021**
Indian Institute of Science (IISc), Bangalore — Cumulative GPA: 9/10
B.S. Mechanical Engineering — **Graduated May 2019**
Birla Institute of Technology and Science (BITS) – Pilani — Cumulative GPA: 9.4/10

RESEARCH EXPERIENCE

PhD Student, Graduate Research Assistant
Theoretical Mechanobiology and Materials *Philadelphia, PA* — **August 2021 – Present**
Advised by Dr. Vivek Shenoy | Developed theoretical frameworks and computational models to investigate engineering and biological system interactions.

Project 1: Extracellular matrix mechanics as a regulator of cancer cell migration

- Developed, independently, a physics-based theoretical model and applied Monte Carlo simulations to investigate cancer cell migration on viscoelastic substrates and synthesized findings into a manuscript submitted to *Nature Physics*.
- Analyzed experimental biological data and various pharmacological interventions to quantify cancer cell migration dynamics, utilizing advanced data analytics and statistical techniques to evaluate parameters such as velocity autocorrelation, migration diffusivity, and track straightness.
- Elucidated, for the first time in the field, the fundamental physics of glassy adhesion dynamics, revealing how the interplay between cellular and extracellular timescales governs cell motility.

Project 2: Mechano-electro-osmotic model for solid stress in cancer spheroids

- Formulated a multiscale physics model integrating intracellular biological processes with tissue-level mechanics to probe the critical mechanisms of solid stress generation in avascular tumors, guiding the design of targeted experiments.
- Ideated, developed, and deployed advanced ML/AI-based segmentation and 3D tracking tools to analyze complex experimental imaging of 3D cancer spheroids, accelerating data analysis workflows by 50% and enabling extraction of critical morphological and pharmacological features of tumor growth.
- Defended research findings in rigorous internal technical reviews, demonstrating analytical rigor and securing approval for manuscript preparation and forthcoming journal submission.

Project 3: T-cell activation and spreading regulated by substrate stress relaxation

- Established a physics-informed model to explore immune synapse formation, focusing on how antigen mechanical properties influence T cell activation and the formation of effective immune synapses for antigen elimination.
- Engineered experiments for T cell activation, cytoskeletal imaging, traction force microscopy, and RNA sequencing analysis to validate theoretical predictions.
- Uncovered novel insights on T cell efficacy regulation by modulating time-dependent properties of antigen-presenting cells or their culture substrates, providing valuable design principles for CAR T cell therapy culturing processes.

Project 4: Deep learning methods for 3D chromatin structure prediction

- Conceptualized AI-driven frameworks leveraging diffusion models, latent space representations, and SE(3) equivariant graph neural networks (GNNs) to predict single-cell Hi-C contact maps and 3D chromatin organization from bulk DNA-seq/ATAC-seq.
- Structured a multi-modal deep learning approach that integrates spatial chromatin architecture with transcriptomics, proteomics, and metabolomic, enabling accurate prediction of gene expression and regulatory interactions.

Graduate Research Assistant
Biomechanics Lab *Bangalore, India* **June 2020 – May 2021**
Advised by Dr. Namrata Gundiah | Master's thesis: Mechanics of bio-inspired fiber reinforced composites.

- Investigated wrinkling instabilities in tissue bi-layers (e.g., skin, brain-cortex interfaces) using custom experimentally fabricated bio-inspired, fiber-reinforced composites made of PDMS matrices.
- Devised analytical models in ABAQUS to predict surface instabilities in bi-layer biological structures and accurately demonstrated stretch dependence of various buckling states in film-substrate systems.
- Developed a haptics-based platform for intravenous catheterization training, simulating real-life mechanical feedback and tactile interface to enhance medical professional training experience.

TECHNICAL EXPERIENCE

Team member
Transcriptomics Case Study on COVID-19 *Philadelphia, PA* **January 2025 – Present**

- Analyzed mechanobiological gene expression alterations in lung cells in response to SARS-CoV-2 infection, aiming to uncover key molecular mechanisms driving disease progression using RNA-seq data from infected samples (GSE147507, ARCHS4).
- Performed differential gene expression analysis and pathway enrichment in R, leveraging DESeq2 for statistical analysis and ggplot2 for data visualization, to investigate ECM remodeling and cytoskeletal dynamics, with implications for therapeutic strategy development.

Pro-bono healthcare consultant
Penn Biotech Group (PBG) *Philadelphia, PA* **September 2023 – January 2024**

- Collaborated with a multidisciplinary team of 8 to analyze the pricing landscape and product launch timeline, compiling results to provide strategic recommendations to a gene-editing biotech startup.
- Reviewed academic research on the scientific product to assess feasibility and the competitive landscape, offering insights on product pricing and refinement of the product concept.

Medical Device Intern
Asian Institute of Gastroenterology *Bangalore, India* **January 2019 – May 2019**

- Designed a novel, low-cost medical device in collaboration with the Asian Institute of Gastroenterology for endoscopy instruments to track movements upon deployment and measure pancreatic pseudo-cyst volume.
- Developed a model prototype using SolidWorks and iterated for sensor integration compatibility.
- Implemented a program to combine positional data coupled with endoscopy image analysis to aid doctors in decision-making for procedural diagnostics.

TEACHING AND LEADERSHIP EXPERIENCE

Graduate Teaching Assistant
University of Pennsylvania *Philadelphia, PA*

- Introduction to Mechanics **August 2023 – December 2023**
- Principles and Techniques of Applied Mathematics **January 2023 – May 2023**
- Introduction to Mechanics **August 2022 – December 2022**

Vice President
Mechanical Engineering Graduate Association *Philadelphia, PA* **October 2022 – November 2023**

- Initiated improvements for graduate student qualifying exam procedures as a representative for mechanical engineering student body.
- Executed 13 social events, uniting more than 100 graduate students, faculty, and staff to enhance community.
- Managed professional development events and mock qualifying exam sessions, aiding junior students in their exam preparation.

TECHNICAL SKILLS

Modeling and Analysis: COMSOL, physics-based models, Monte Carlo, molecular dynamics, Blender, SolidWorks, AutoCAD, Creo, Ansys, ABAQUS
Programming: Python, MATLAB, C++, C, R, SQL, Arduino IDE, Maple
Miscellaneous: Git, Visual Studio, Microsoft Office Suite, GraphPad Prism, Inkscape, Image J

RELEVANT COURSES AND CERTIFICATION

Courses:

1. Introduction to Computational Biology
2. Data-driven Modeling and Probabilistic Scientific Computing
3. AI4Science/Science4AI
4. Multiscale Modeling of Chemical and Biological Systems
5. Atomistic Modeling in Materials Science
6. DIY Transcriptomics

Certifications:

1. Certificate of Advanced Scientific Computing — **Penn Institute of Computational Science (PICS)**
2. Supervised Machine Learning — **Coursera**
3. Deep Learning with PyTorch — **DataCamp**
4. Bioinformatics Specialization (In progress) — **Coursera**

HONORS AND AWARDS

1. Outstanding Teaching Assistant (University of Pennsylvania) — **June 2024**
2. Graduated with distinction (Indian Institute of Science) — **August 2021**
3. All India Rank – 387/170,000 (Graduate Aptitude Test in Engineering – GATE) — **May 2019**
4. Graduated with distinction (Birla Institute of Technology and Science) — **May 2019**
5. Merit Scholarship (Birla Institute of Technology and Science) — **August 2015 – May 2019**

PUBLICATIONS

1. (In preparation)
   Title: T cell spreading and actomyosin dynamics in immune synapse formation governed by substrate relaxation.
2. (In preparation)
   Title: Hydraulics and ion accumulation from metabolite deprivation drive solid stress in tumor spheroids.
3. Nature Physics (Submitted)
   Title: Glassy adhesion dynamics govern transitions between sub- and super-diffusive cell migration on viscoelastic substrates.

CONFERENCES

1. Biophysical Society (BPS) Meeting, February 2024 – Philadelphia, PA
   Poster: Substrate stress relaxation governs the transition from sub-diffusive to super-diffusive migration.
2. Biomedical Engineering Society (BMES), October 2023 – Seattle, WA
   Presenter: Substrate stress relaxation governs the transition from sub-diffusive to super-diffusive migration.
3. Big Data in Biomedical and Population Health Sciences, September 2023 – Philadelphia, PA
   Attendee
4. Annual Mechanobiology Symposium, April 2023 – Philadelphia, PA
   Poster: Chemo-mechanical diffusion waves explain collective dynamics of immune cell podosomes.
5. American Physical Society, March 2023 – Las Vegas, NV
   Presenter: Cell migration on viscoelastic substrates.